

Relevance-Based Prediction: Factor Selection for Saugus Schools

Utilizing Czasonis, Kritzman & Turkington (2024) — Applied to MA School District Analysis

Four outcomes: MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, Education Budget Share

RBP run once with ALL candidates — Exhibit 5 importance selects the lean feature set

Features with positive importance kept; ≤ 0 importance pruned (adds noise, not signal)

Saugus analyzed as prediction task; most/least relevant towns identified per Exhibit 4

Leave-one-out LOO r validates lean feature set across all MA districts

Selected Feature Counts			
MCAS Grades 3-8	Dropout Rate	MCAS Grade 10 (ELA)	Education Budget Share
Candidates: 17 features	Candidates: 18 features	Candidates: 18 features	Candidates: 18 features
Lean (imp > 0): 14 features	Lean (imp > 0): 15 features	Lean (imp > 0): 14 features	Lean (imp > 0): 17 features
LOO r = +0.946	LOO r = +0.837	LOO r = +0.902	LOO r = +0.903

Pre-Specified Candidate Pool — Feature Reduction

Features selected on domain grounds before running RBP. Reduces K/N ratio from ~0.19 to ~0.09 (target ≈ 0.085).

Kept — 14 pure predictors

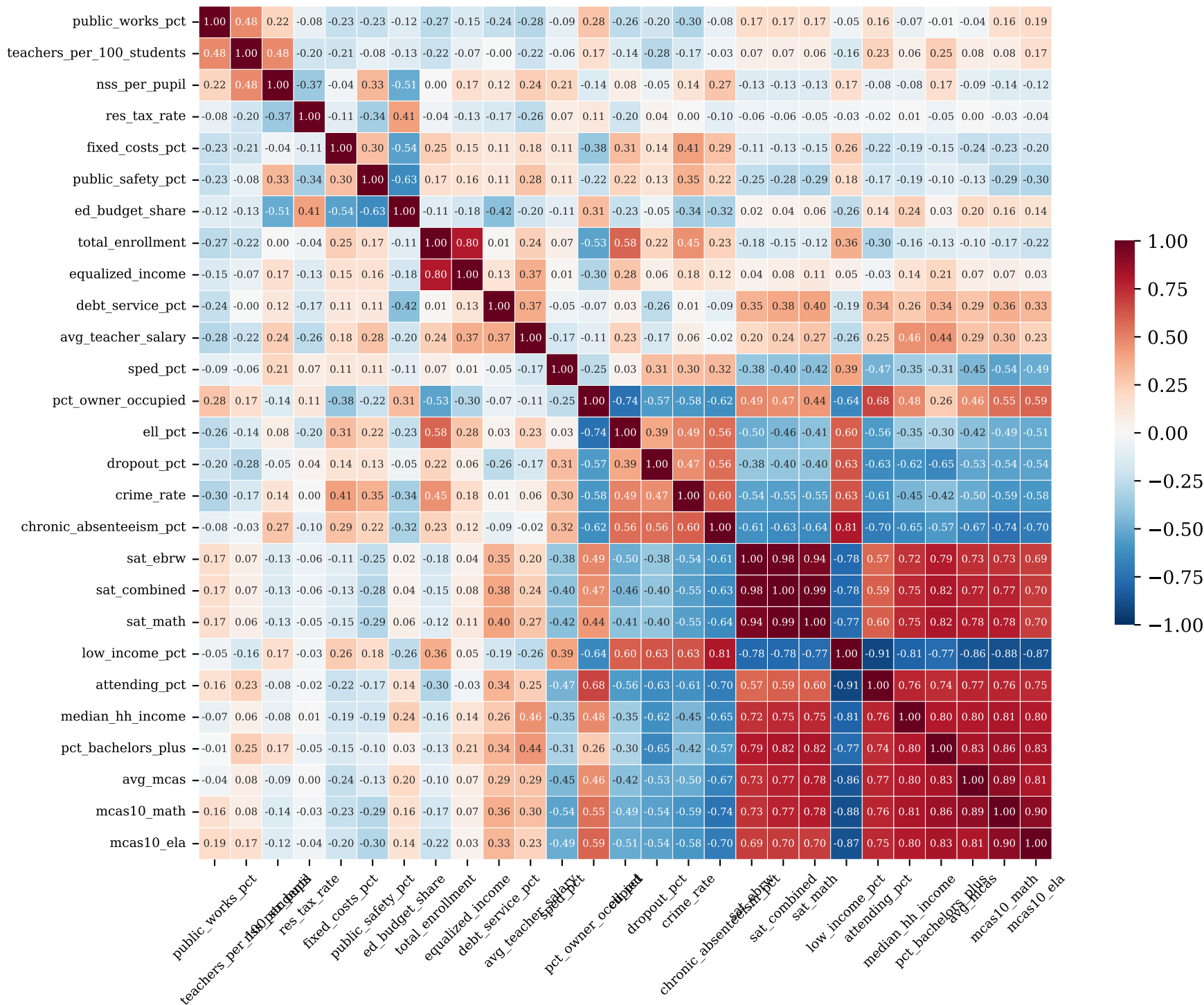
Group	Feature	Measures
Poverty	low_income_pct	% low-income students
Wealth	median_hh_income	Median household income
Wealth	equalized_income	Equalized property value/capita
Human capital	pct_bachelors_plus	% adults with bachelor's+
Housing	pct_owner_occupied	% owner-occupied homes
Community	crime_rate	Crime incidents per capita
Community	res_tax_rate	Residential tax rate
Engagement	chron_absenteeism_pct	% chronically absent
Demographics	ell_pct	% English language learners
Demographics	sped_pct	% special education students
Size	total_enrollment	District enrollment
Staffing	teachers/100_students	Teachers per 100 students
Staffing	avg_teacher_salary	Average teacher salary
Spending	nss_per_pupil	Net school spending/pupil

Dropped — 12 redundant/problematic features

Dropped	Kept instead	Reason
high_needs_pct	low_income_pct	$r=+0.981$ — near-duplicate
acs_poverty_pct	low_income_pct	$r=+0.87$ — redundant poverty
foundation_budget_pp	nss_per_pupil	$r=+0.92$ — redundant spending
ch70_per_pupil	equalized_income	$r\approx-0.87$ — inverse wealth proxy
teacher_fte	total_enrollment	$r=+0.994$ — linear with enroll
total_population	total_enrollment	$r=+0.953$ — size proxy
teachers_per_100_fte	t_per_100_students	$r=+0.872$ — duplicate ratio
teacher_spending_pp	nss_per_pupil	$r\approx+0.85$ — redundant spending
gf_exp_per_capita	res_tax_rate	$r\approx+0.75$ — redundant municipal
pct_65_plus	—	Weak signal, no policy lever
com_tax_rate	res_tax_rate	$r\approx+0.65$ — commercial tax dup
mcas10_math	mcas10_elar	$r=+0.90$ — one grade-10 measure

Correlation threshold for deduplication: $|r| > 0.87$. Cross-outcome features (avg_mcas, mcas10_ela, dropout_pct, attending_pct) handled separately per model via exclusion rules.

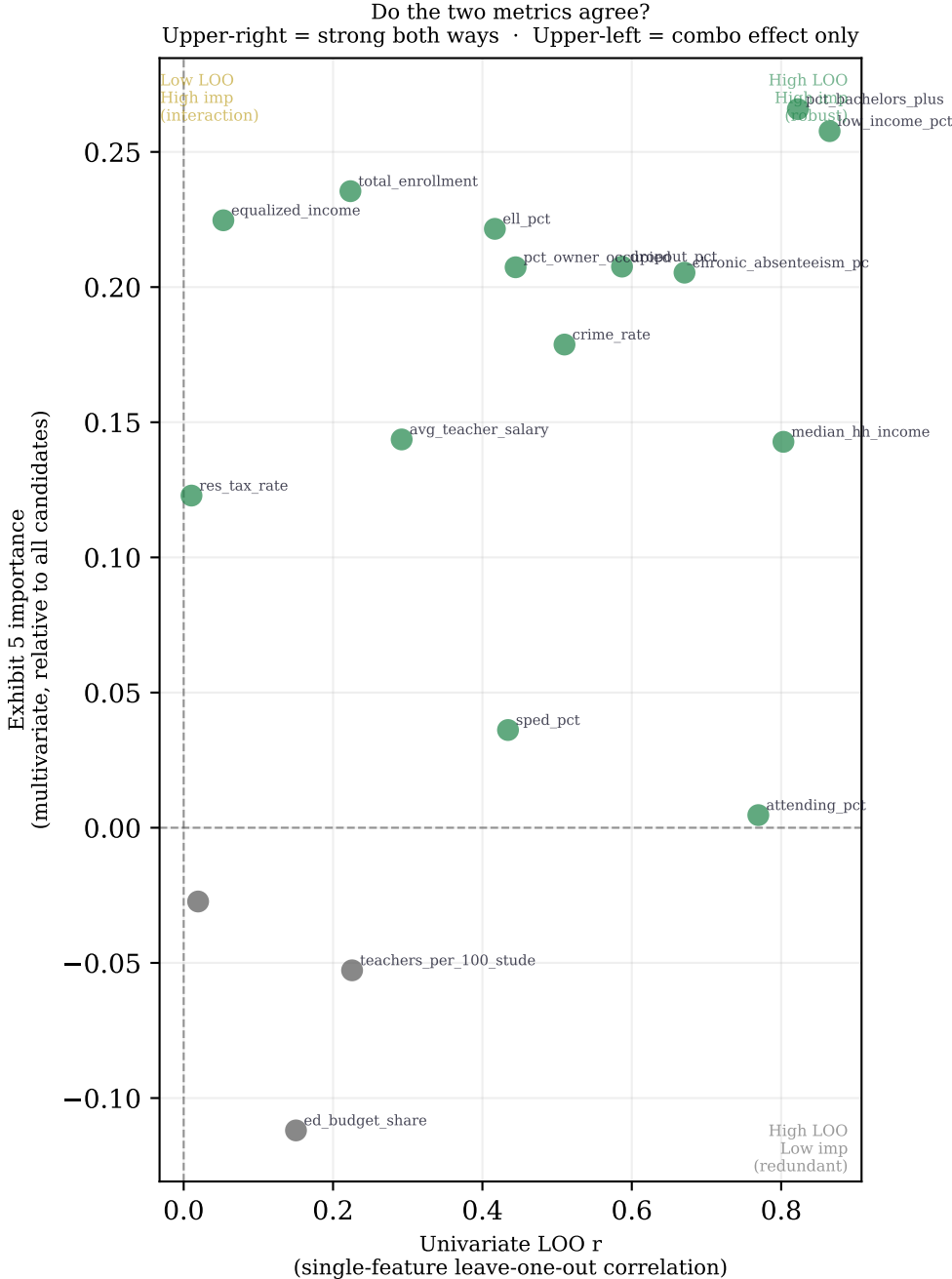
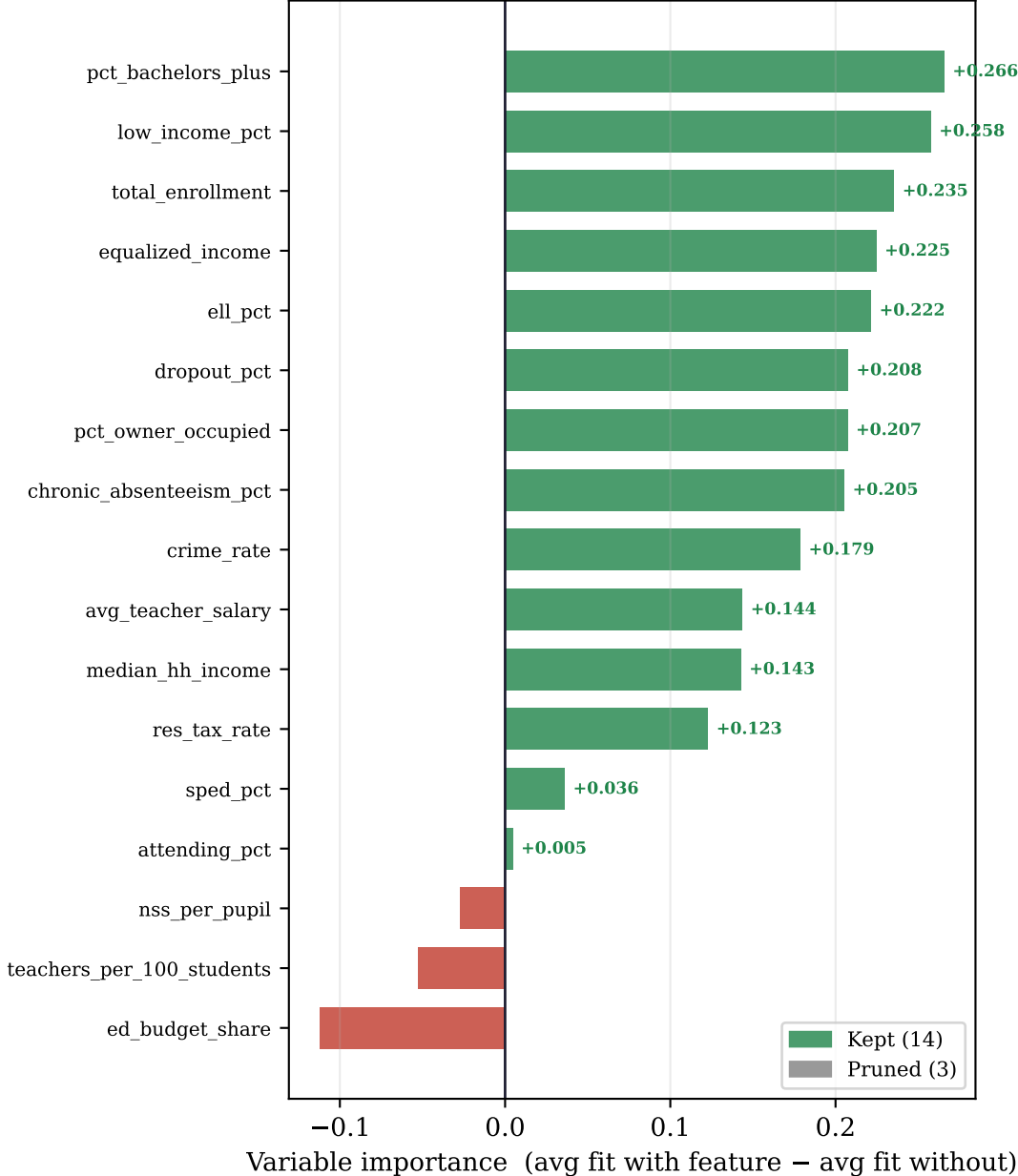
Feature Correlation Matrix — All Candidates (hierarchically clustered)
Features ordered by |correlation| — dark red/blue clusters = high redundancy



Factor Selection via Variable Importance: MCAS Grades 3-8

17 candidates · n_random=1000 · 14 features kept (importance > 0) · 3 pruned

All Candidates — Variable Importance
Green = kept (importance > 0) · Grey/red = pruned

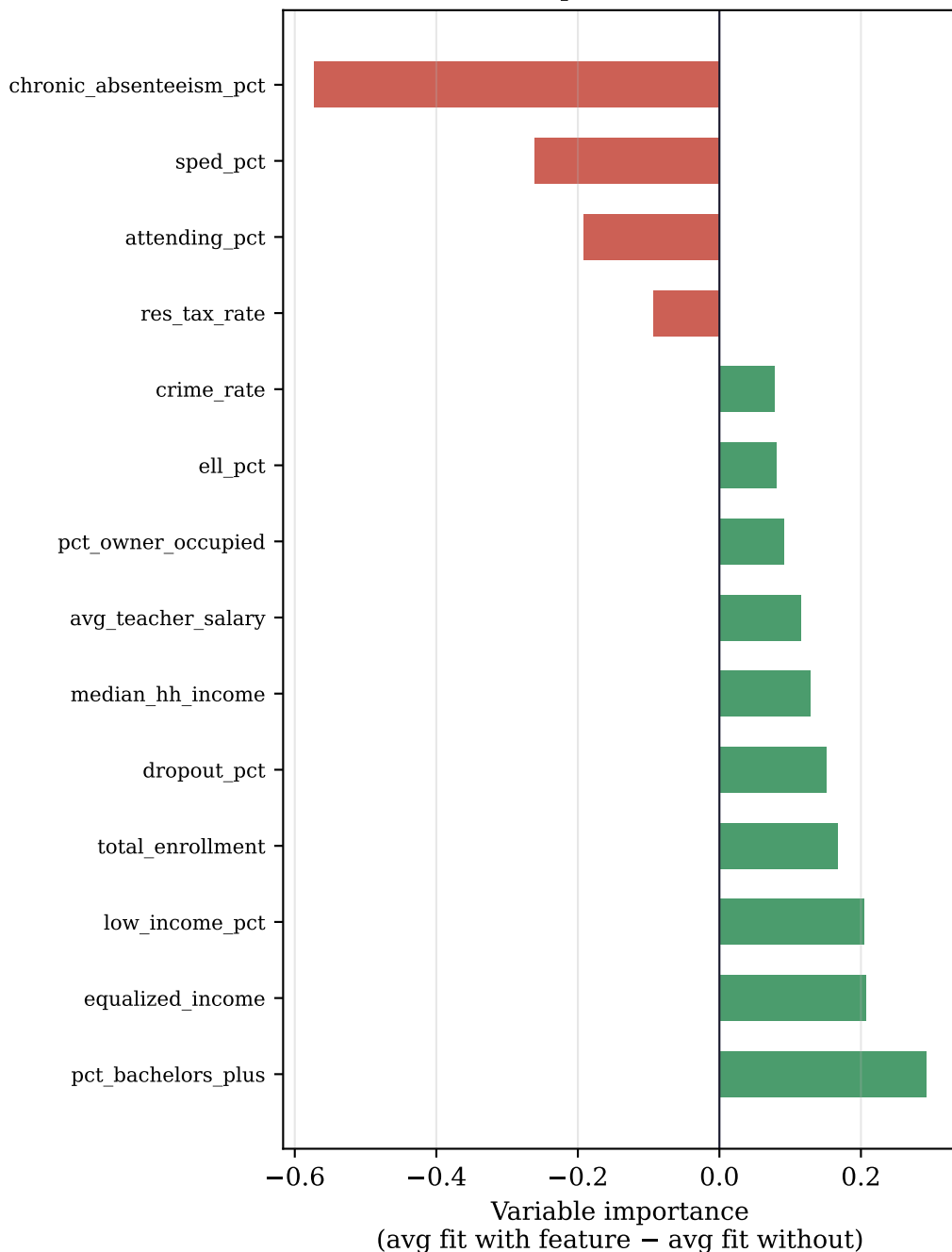


Variable importance: avg adjusted fit of grid cells containing feature – avg fit of cells without it. Univariate LOO r: single-feature leave-one-out correlation.

Saugus RBP Analysis: MCAS Grades 3-8

Prediction task = Saugus · Predicted: 28.7pp Actual: 29.0pp Gap: +0.3pp (Fit: 0.2701)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	avg_mcas
Randolph	0.0377	0.26
Everett	0.0369	0.18
Pittsfield	0.0354	0.24
Agawam	0.0318	0.34
Revere	0.0310	0.23

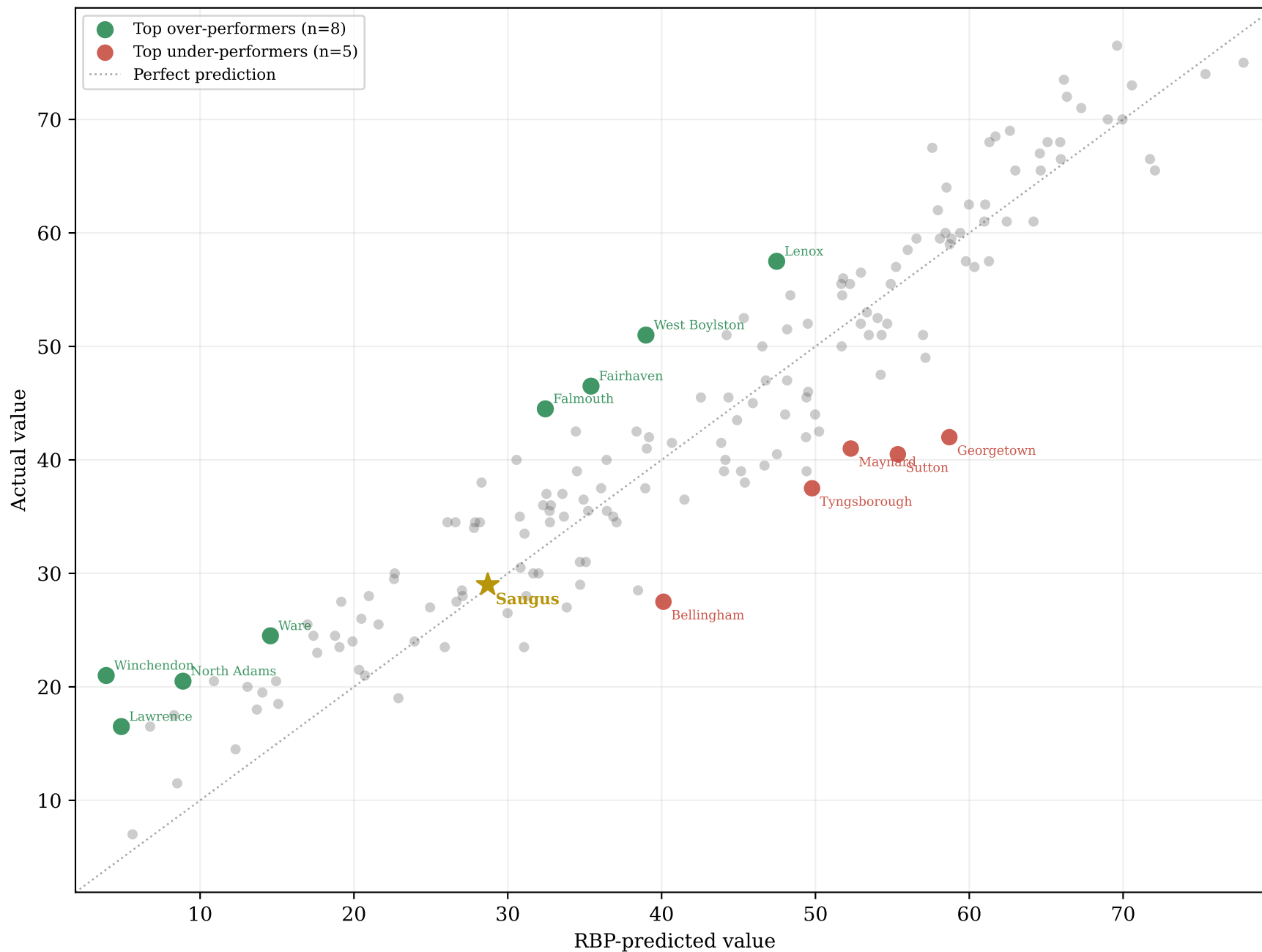
Least Relevant Towns (lowest weight)

Town	Weight	avg_mcas
Reading	-0.0061	0.62
Ipswich	-0.0059	0.56
Natick	-0.0057	0.62
Andover	-0.0057	0.66
Foxborough	-0.0052	0.51

Over- and Under-Performers: MCAS Grades 3-8

Saugus gap: +0.3pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — MCAS Grades 3-8 (LOO r = 0.946)



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: MCAS Grades 3-8

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 14 shown

Feature	Saugus	Winchendon	Falmouth	West Boylston	Imp
chronic_absenteeism_pct	31.2	35.6	24.0	15.6	-0.573
pct_bachelors_plus	31.4	23.9	51.9	41.1	+0.293
sped_pct	18.8	27.9	24.4	19.5	-0.261
equalized_income	1.17B	278M	1.68B	309M	+0.207
low_income_pct	46.6	56.8	39.5	31.5	+0.204
attending_pct	64.4	36.4	68.0	72.9	-0.191
total_enrollment	3,084	1,360	3,130	986.0	+0.168
dropout_pct	2.90	7.50	1.00	0.40	+0.151
median_hh_income	100,819	85,160	92,862	99,674	+0.129
avg_teacher_salary	89,778	83,722	92,617	89,252	+0.115
res_tax_rate	10.7	12.5	6.28	14.8	-0.094
pct_owner_occupied	73.9	77.8	82.7	79.6	+0.092
ell_pct	13.0	1.90	7.50	6.60	+0.081
crime_rate	2,813	3,287	3,500	1,112	+0.078

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	29.0	28.7	+0.3pp
Winchendon	21.0	3.9	+17.1pp
Falmouth	44.5	32.4	+12.1pp
West Boylston	51.0	39.0	+12.0pp
Lawrence	16.5	4.9	+11.6pp
North Adams	20.5	8.9	+11.6pp
Fairhaven	46.5	35.4	+11.1pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grades 3-8

Predicted 28.7pp · Actual 29.0pp · Gap +0.3pp (higher is better) · Saugus ranks 103 of 169 MA districts on this measure

The Bottom Line

Saugus scores 29.0% on MCAS grades 3-8 (ELA + Math, meeting/exceeding), almost exactly its demographic prediction of 28.7% — a +0.3pp gap that is essentially on target. Saugus ranks 103rd of 169 MA districts, around the middle of the pack. This is the “baseline” result: a district with Saugus's income, poverty, and enrollment profile is expected to land here, and it does.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	-0.57
Adults with a bachelor's degree+	31.4%	46.9%	+0.29
Special education students	18.8%	19.8%	-0.26
Equalized property value per capita	\$1.17B	\$935M	+0.21
% students from low-income families	46.6%	30.4%	+0.20

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	25.1%	-6.1pp
Adults with a bachelor's degree+	31.4%	28.7%	-2.7pp
Special education students	18.8%	20.6%	+1.8pp
Equalized property value per capita	\$1.17B	\$309M	-861M
% students from low-income families	46.6%	48.1%	+1.5pp

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

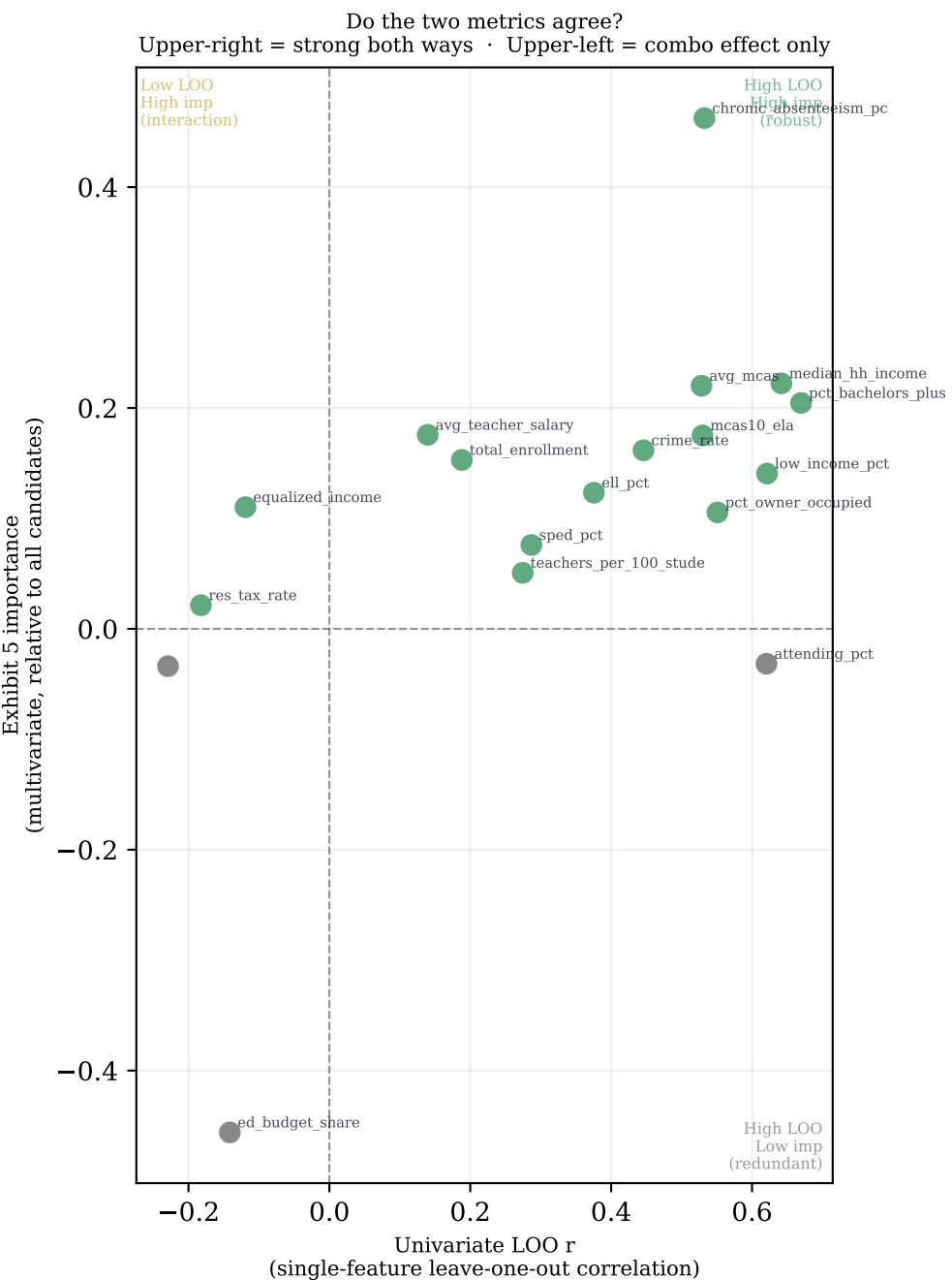
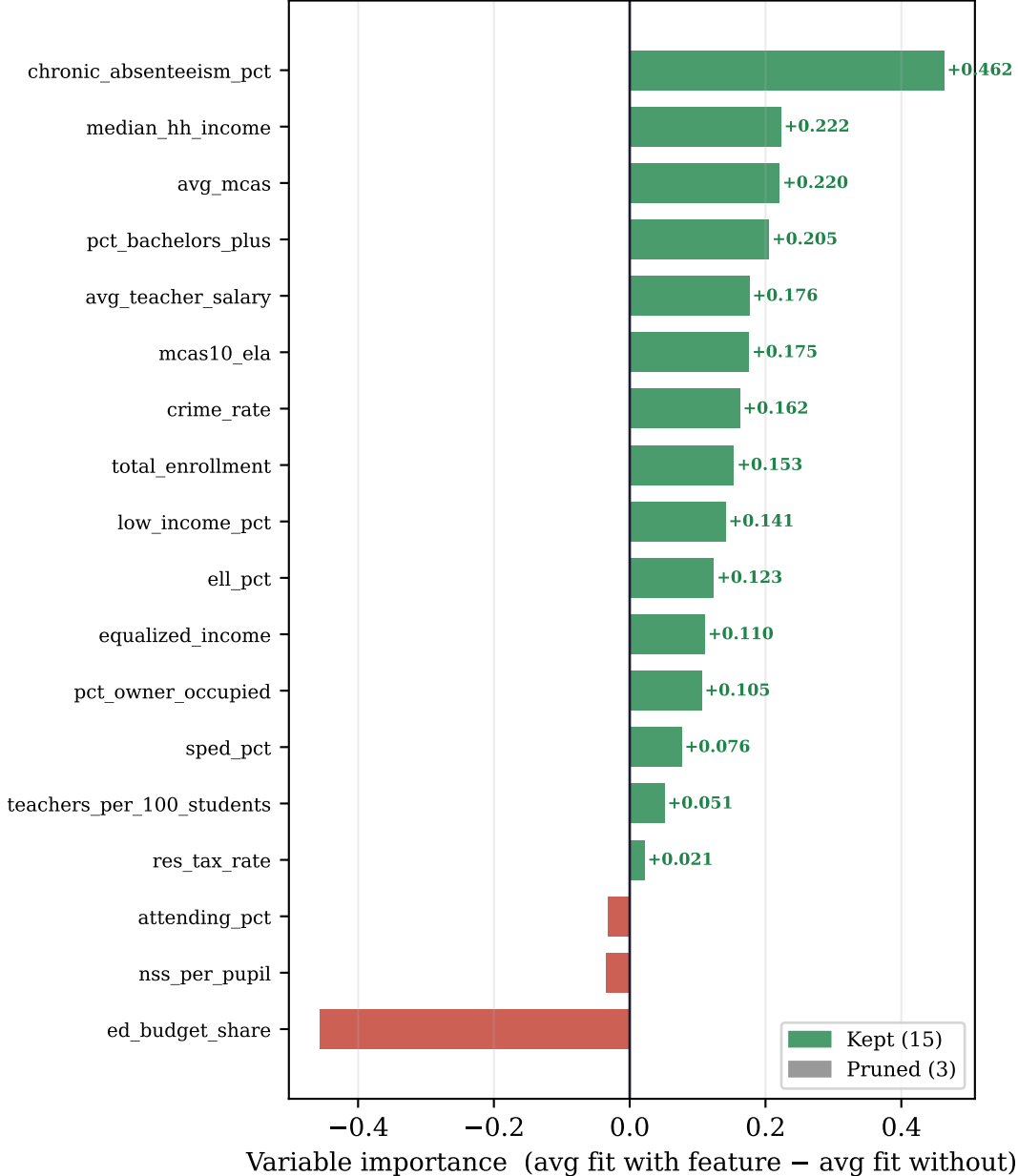
The Takeaway

Because Saugus is on target here, the importance scores point less to “what's wrong” and more to “what would move the needle.” Chronic absenteeism (31.2%, nearly double the 15.8% state median) is the single largest local driver — and the largest gap versus over-performing districts. Closing even part of that gap is the most plausible route from an average result to an above-average one.

Factor Selection via Variable Importance: Dropout Rate

18 candidates · n_random=1000 · 15 features kept (importance > 0) · 3 pruned

All Candidates — Variable Importance
Green = kept (importance > 0) · Grey/red = pruned

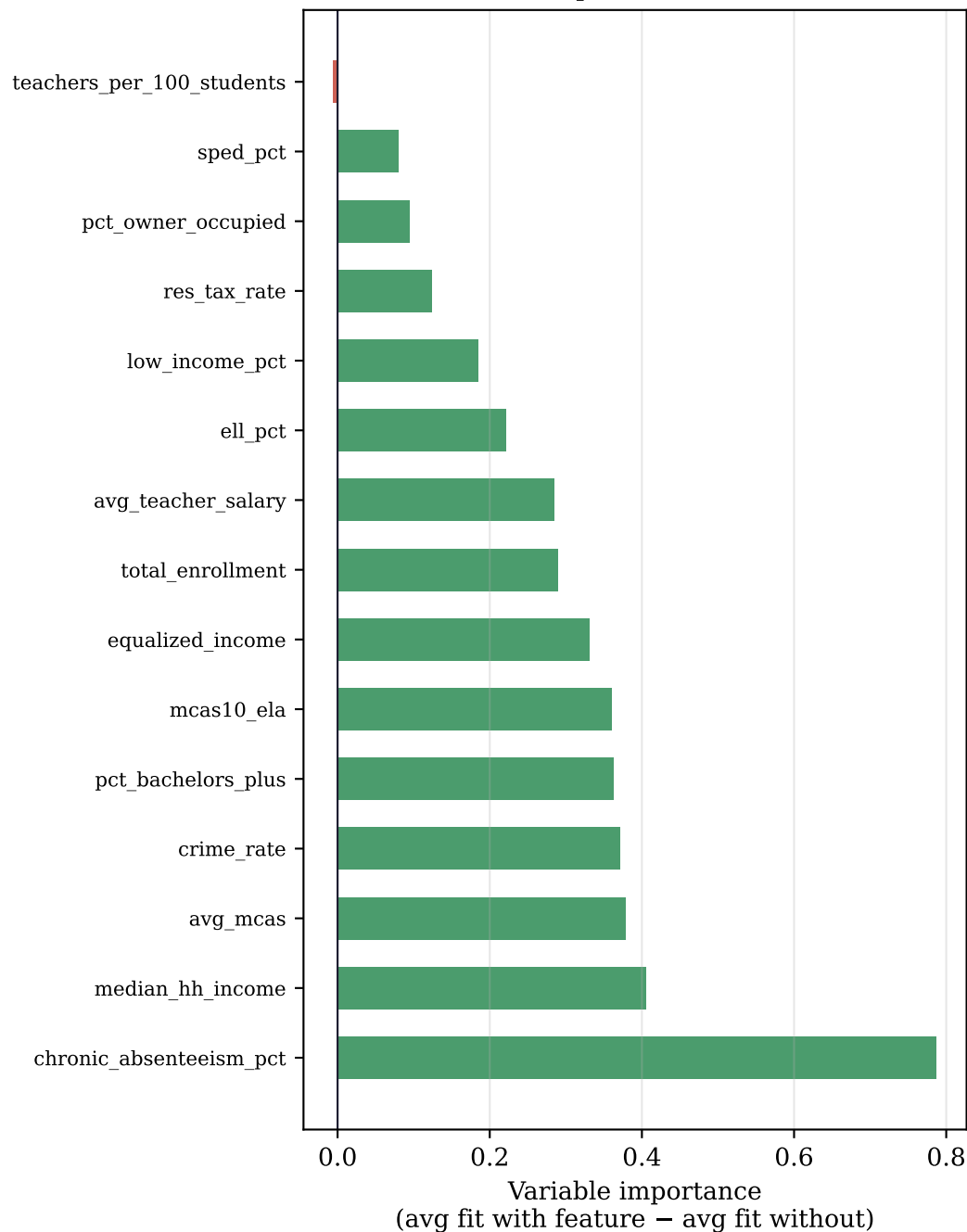


Variable importance: avg adjusted fit of grid cells containing feature – avg fit of cells without it. Univariate LOO r: single-feature leave-one-out correlation.

Saugus RBP Analysis: Dropout Rate

Prediction task = Saugus · Predicted: 3.7pp Actual: 2.9pp Gap: -0.8pp (Fit: 0.2618)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	dropout_pct
Winchendon	0.0747	7.50
North Brookfield	0.0554	7.40
Randolph	0.0448	3.30
Dracut	0.0443	1.40
Everett	0.0442	5.20

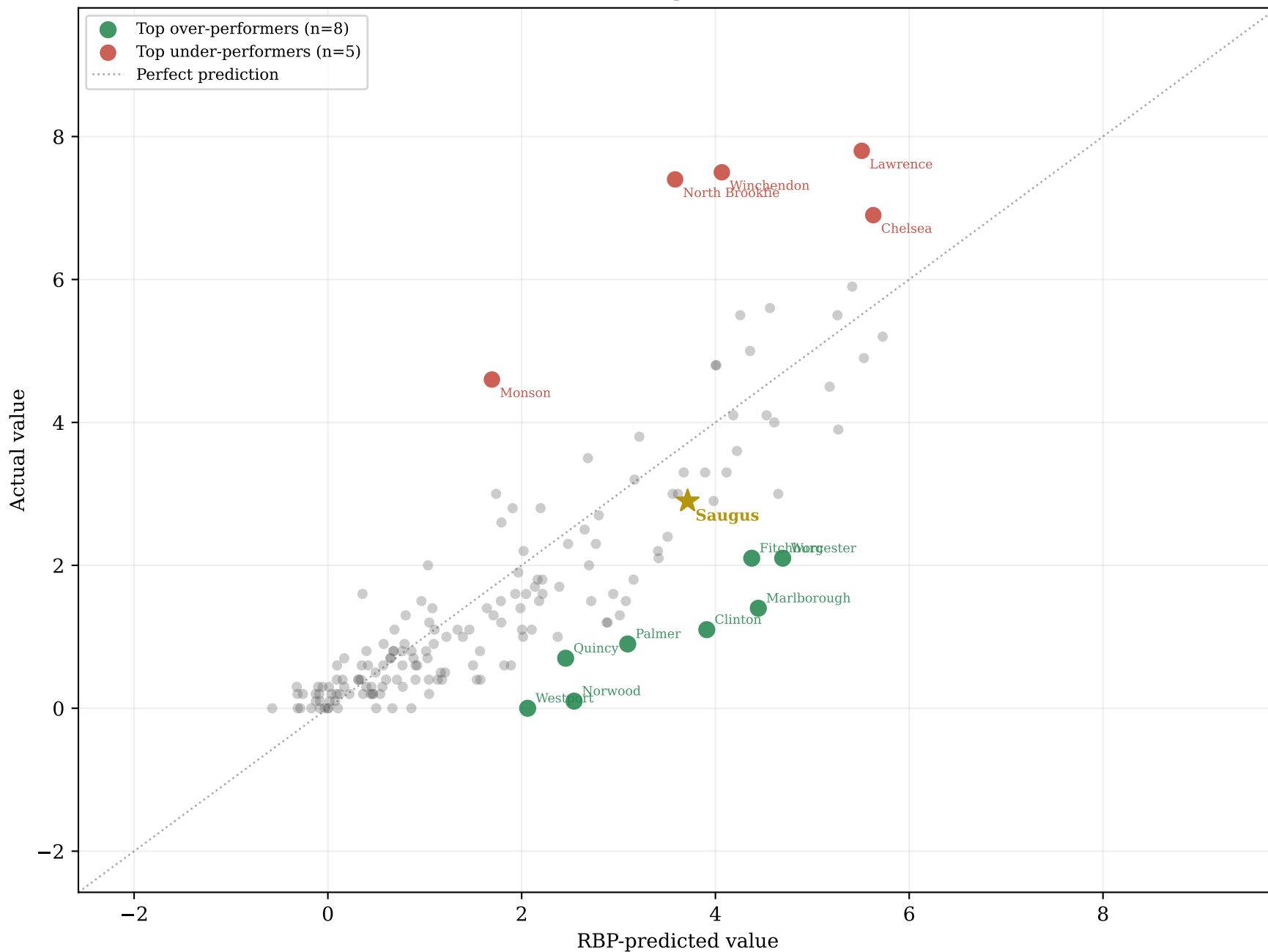
Least Relevant Towns (lowest weight)

Town	Weight	dropout_pct
Mansfield	-0.0112	0.20
Dedham	-0.0112	0.40
Natick	-0.0109	0.70
Andover	-0.0107	0.20
Bedford	-0.0103	0.20

Over- and Under-Performers: Dropout Rate

Saugus gap: -0.8pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — Dropout Rate (LOO r = 0.837)



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: Dropout Rate

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Marlborou	Clinton	Worcester	Imp
chronic_absenteeism_pct	31.2	23.5	7.70	28.3	+0.787
median_hh_income	100,819	95,047	93,849	67,544	+0.405
avg_mcas	0.29	0.24	0.30	0.23	+0.378
crime_rate	2,813	3,695	324.5	5,437	+0.371
pct_bachelors_plus	31.4	40.6	35.1	34.1	+0.362
mcas10_ela	0.47	0.40	0.43	0.40	+0.360
equalized_income	1.17B	1.65B	514M	5.00B	+0.330
total_enrollment	3,084	5,572	2,237	27,606	+0.289
avg_teacher_salary	89,778	87,213	87,059	101,529	+0.284
ell_pct	13.0	32.9	19.7	30.4	+0.221
low_income_pct	46.6	58.3	52.7	71.4	+0.185
res_tax_rate	10.7	10.2	13.1	13.8	+0.124
pct_owner_occupied	73.9	56.2	52.9	42.4	+0.095
sped_pct	18.8	16.4	21.9	21.9	+0.080
teachers_per_100_student	5.87	7.36	7.31	7.40	-0.005

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	2.9	3.7	-0.8pp
Marlborough	1.4	4.4	+3.0pp better
Clinton	1.1	3.9	+2.8pp better
Worcester	2.1	4.7	+2.6pp better
Norwood	0.1	2.5	+2.4pp better
Fitchburg	2.1	4.4	+2.3pp better
Palmer	0.9	3.1	+2.2pp better

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Dropout Rate

Predicted 3.7pp · Actual 2.9pp · Gap -0.8pp (lower is better) · Saugus ranks 34 of 169 MA districts on this measure

The Bottom Line

Saugus's dropout rate is 2.9%, a full 0.8pp better than its predicted 3.7% (lower is better for this measure) — ranking 34th of 169 districts, in the top 20%. Of the four outcomes, this is Saugus's clearest over-performance relative to demographic expectation.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.79
Median household income	\$100,819	\$109,970	+0.40
MCAS 3–8 (% meeting/exceeding)	29.0%	40.0%	+0.38
Crime incidents per 100k residents	2,813	2,241	+0.37
Adults with a bachelor's degree+	31.4%	46.9%	+0.36

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	21.4%	-9.8pp
Median household income	\$100,819	\$94,448	-6,371
MCAS 3–8 (% meeting/exceeding)	29.0%	28.8%	-0.3pp
Crime incidents per 100k residents	2,813	3,684	+871
Adults with a bachelor's degree+	31.4%	34.6%	+3.1pp

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

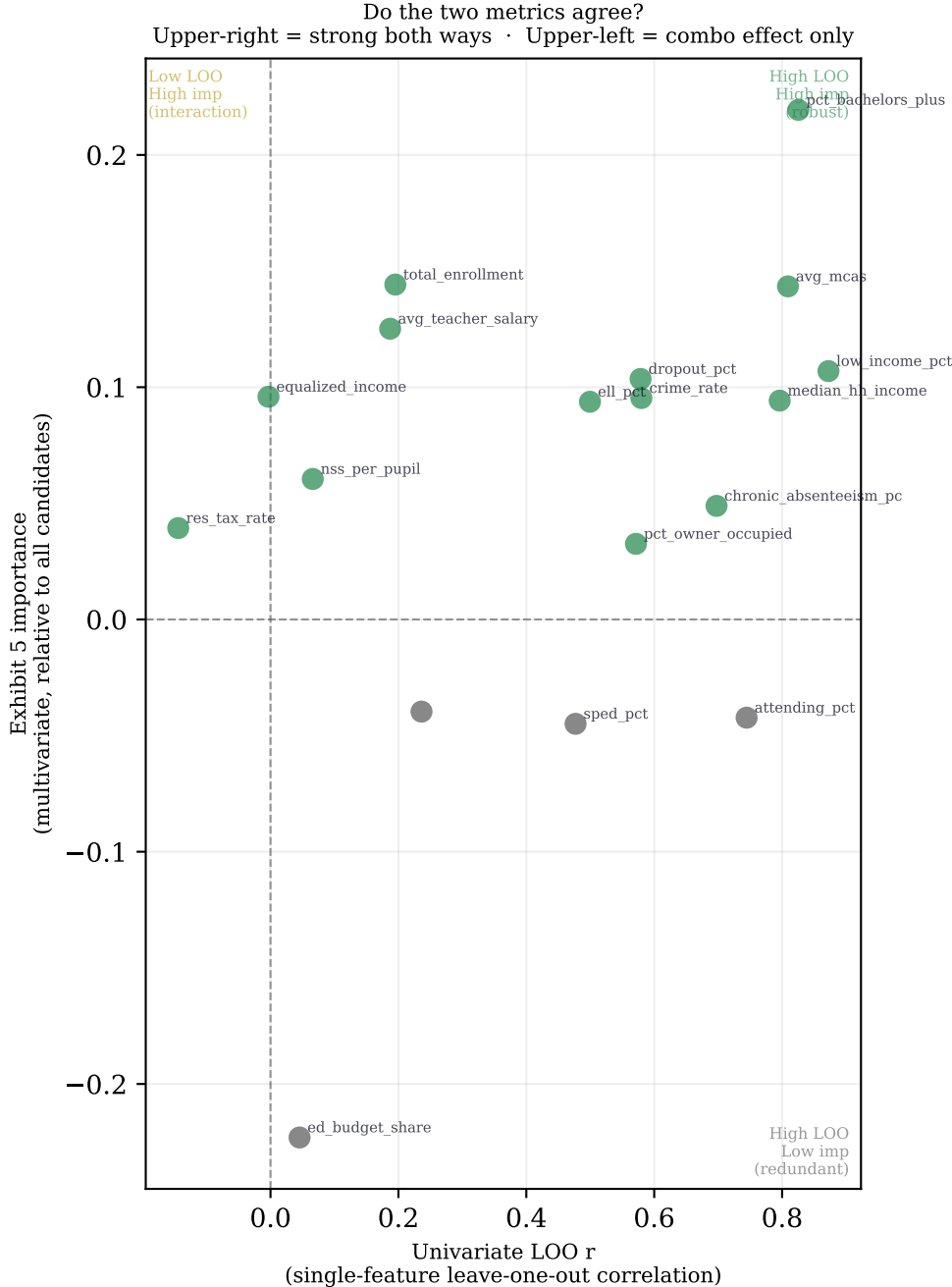
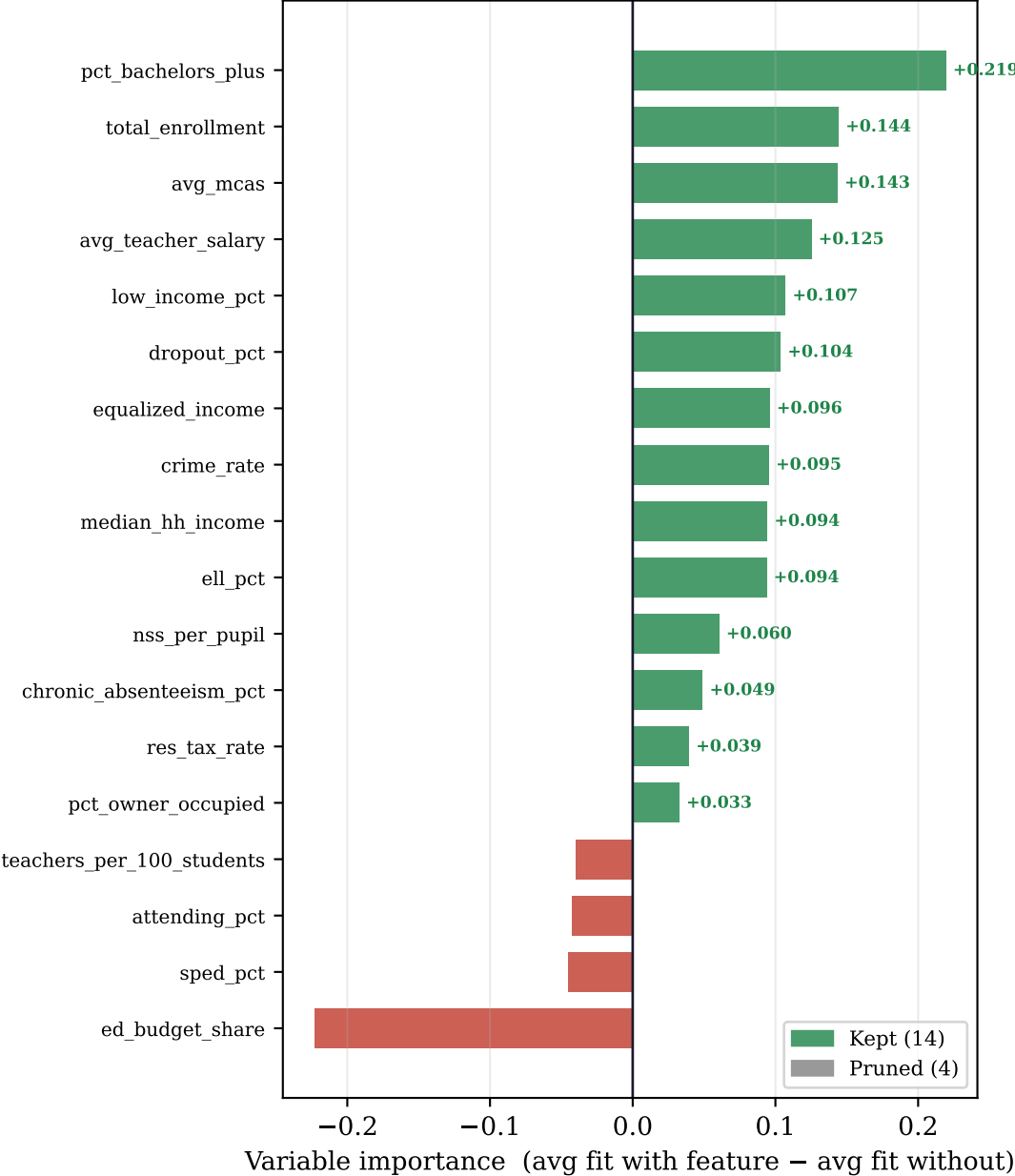
The Takeaway

Chronic absenteeism is overwhelmingly the strongest local driver here — more than double any other factor — yet Saugus still keeps fewer students from dropping out than a district with this absenteeism rate typically would. That gap between “high absenteeism” and “low dropout” suggests Saugus's re-engagement and attendance-recovery efforts are working — and that rising absenteeism is a real risk to this result if it isn't addressed.

Factor Selection via Variable Importance: MCAS Grade 10 (ELA)

18 candidates · n_random=1000 · 14 features kept (importance > 0) · 4 pruned

All Candidates — Variable Importance
Green = kept (importance > 0) · Grey/red = pruned

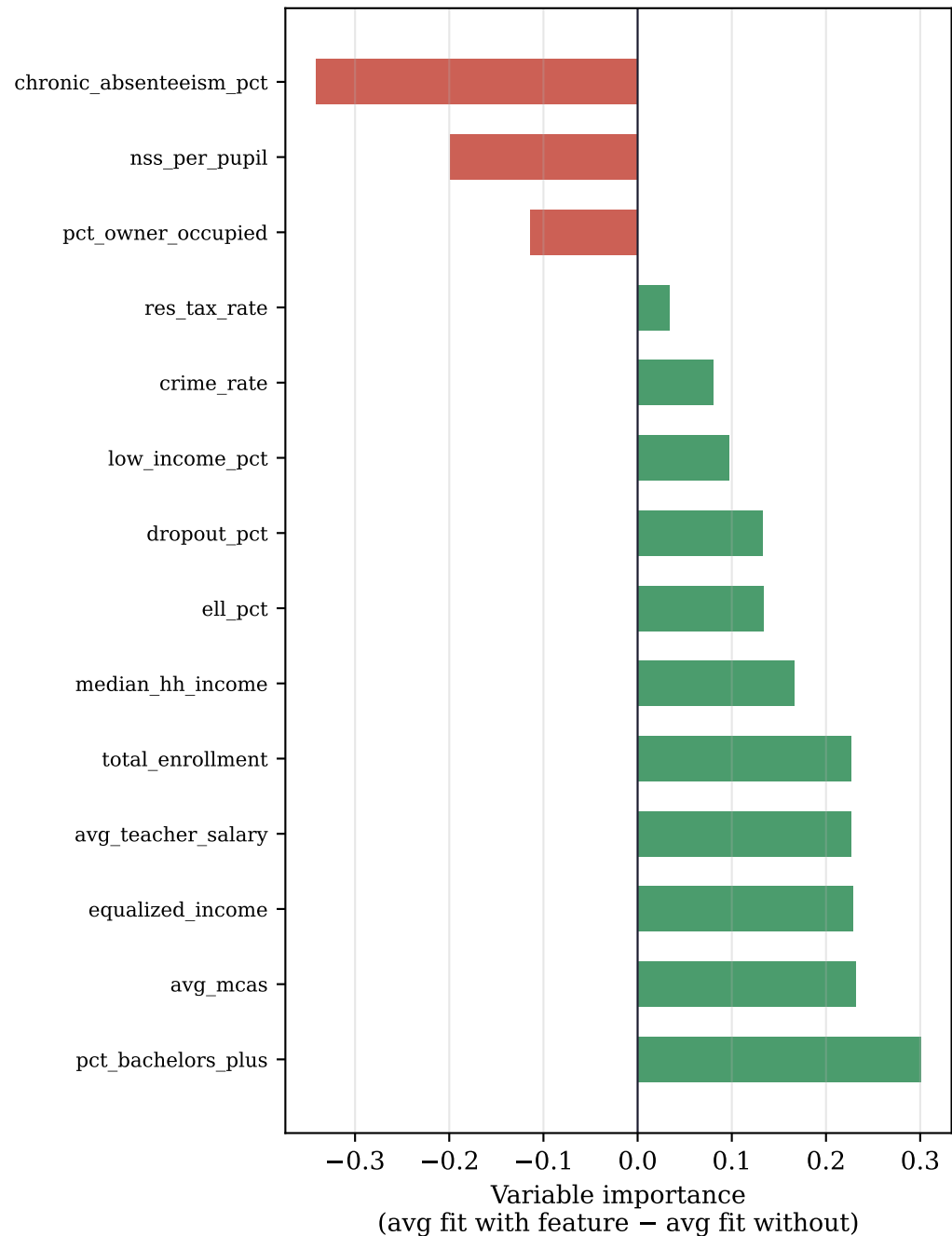


Variable importance: avg adjusted fit of grid cells containing feature – avg fit of cells without it. Univariate LOO r: single-feature leave-one-out correlation.

Saugus RBP Analysis: MCAS Grade 10 (ELA)

Prediction task = Saugus · Predicted: 42.3pp Actual: 47.0pp Gap: +4.7pp (Fit: 0.2643)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	mcas10_ela
Winchendon	0.0533	0.33
Webster	0.0377	0.28
Methuen	0.0363	0.41
Randolph	0.0354	0.36
Everett	0.0343	0.37

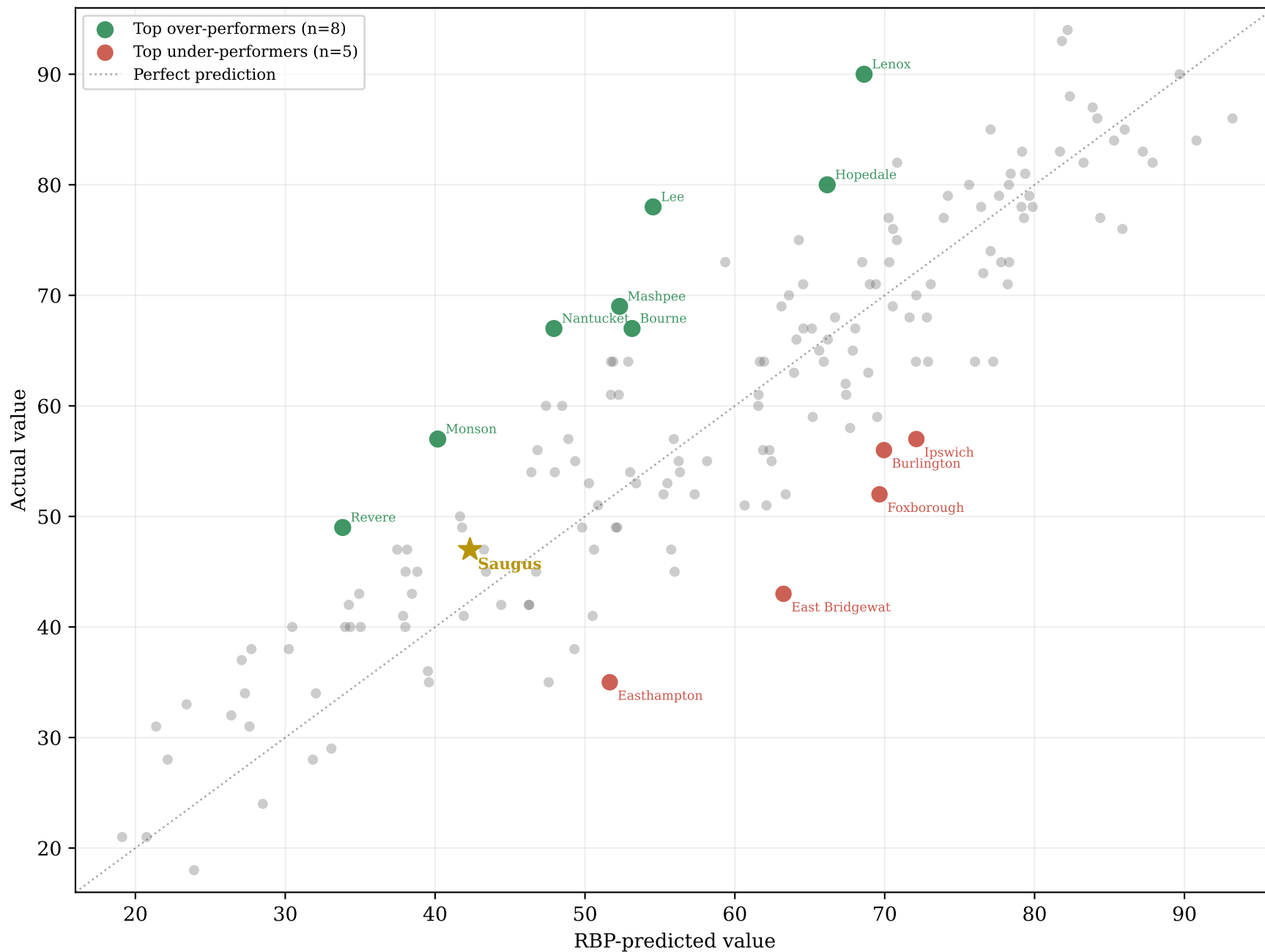
Least Relevant Towns (lowest weight)

Town	Weight	mcas10_ela
Mansfield	-0.0081	0.73
Andover	-0.0080	0.79
Foxborough	-0.0076	0.52
Cohasset	-0.0076	0.93
Bedford	-0.0075	0.83

Over- and Under-Performers: MCAS Grade 10 (ELA)

Saugus gap: +4.7pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — MCAS Grade 10 (ELA) (LOO r = 0.902)



Residual = actual – predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: MCAS Grade 10 (ELA)

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 14 shown

Feature	Saugus	Lee	Lenox	Nantucket	Imp
chronic_absenteeism_pct	31.2	13.3	18.3	25.5	-0.342
pct_bachelors_plus	31.4	45.1	49.5	58.4	+0.301
avg_mcas	0.29	0.34	0.57	0.23	+0.232
equalized_income	1.17B	210M	310M	1.02B	+0.228
avg_teacher_salary	89,778	82,282	84,669	102,575	+0.227
total_enrollment	3,084	746.0	814.0	1,693	+0.227
nss_per_pupil	17,369	24,155	23,945	28,994	-0.200
median_hh_income	100,819	74,222	97,569	119,750	+0.167
ell_pct	13.0	5.60	4.10	19.0	+0.134
dropout_pct	2.90	1.00	0.80	1.50	+0.133
pct_owner_occupied	73.9	78.4	71.3	76.5	-0.115
low_income_pct	46.6	41.2	24.3	34.6	+0.097
crime_rate	2,813	3,348	2,695	3,827	+0.080
res_tax_rate	10.7	11.2	9.07	3.13	+0.034

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	47.0	42.3	+4.7pp
Lee	78.0	54.5	+23.5pp
Lenox	90.0	68.6	+21.4pp
Nantucket	67.0	47.9	+19.1pp
Monson	57.0	40.2	+16.8pp
Mashpee	69.0	52.3	+16.7pp
Revere	49.0	33.8	+15.2pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grade 10 (ELA)

Predicted 42.3pp · Actual 47.0pp · Gap +4.7pp (higher is better) · Saugus ranks 54 of 169 MA districts on this measure

The Bottom Line

Saugus's grade 10 ELA proficiency is 47.0%, 4.7pp above its predicted 42.3% — ranking 54th of 169 districts, in the top third. This is Saugus's strongest relative academic result of the four outcomes.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	-0.34
Adults with a bachelor's degree+	31.4%	46.9%	+0.30
MCAS 3-8 (% meeting/exceeding)	29.0%	40.0%	+0.23
Equalized property value per capita	\$1.17B	\$935M	+0.23
Average teacher salary	\$89,778	\$90,677	+0.23

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	18.1%	-13.1pp
Adults with a bachelor's degree+	31.4%	45.7%	+14.3pp
MCAS 3-8 (% meeting/exceeding)	29.0%	30.2%	+1.2pp
Equalized property value per capita	\$1.17B	\$529M	-642M
Average teacher salary	\$89,778	\$86,260	-3,518

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

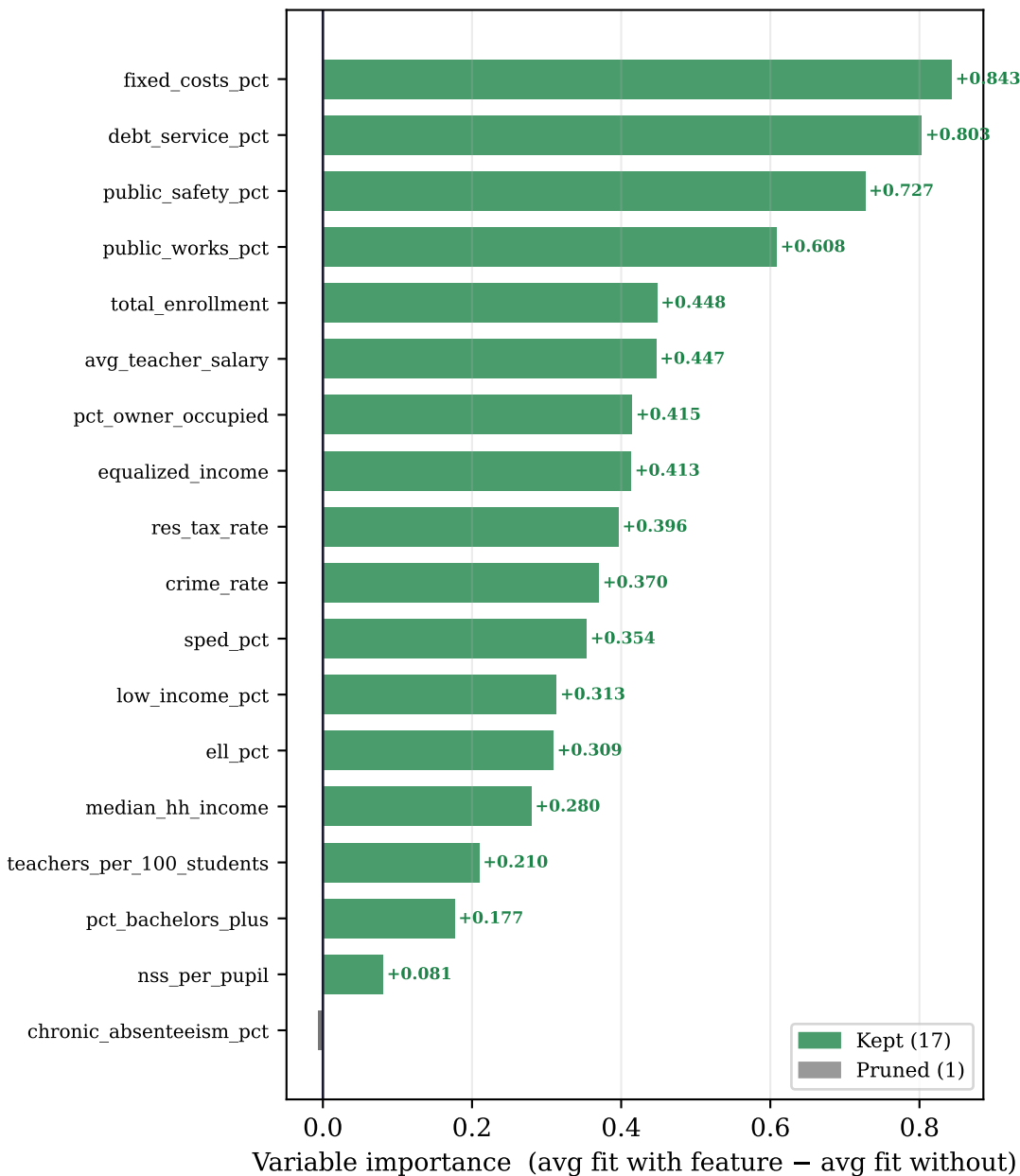
The Takeaway

Chronic absenteeism is again a top driver, and Saugus's rate (31.2%) remains far above the towns it's compared with here. Saugus is overcoming a real headwind on this measure — its grades 3-8 academic base, teacher compensation, and enrollment scale appear to be the offsetting strengths that let it outperform despite high absenteeism.

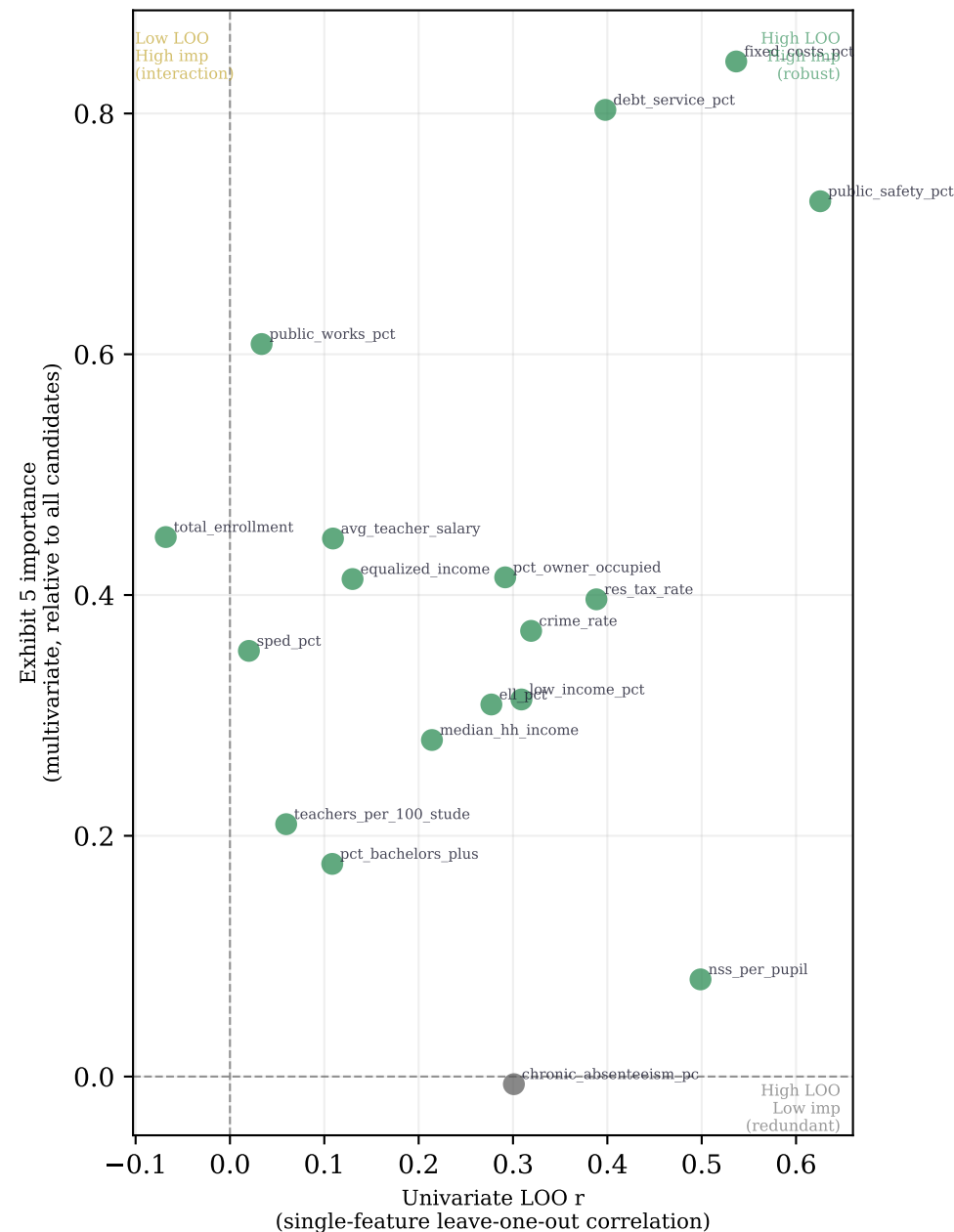
Factor Selection via Variable Importance: Education Budget Share

18 candidates · n_random=1000 · 17 features kept (importance > 0) · 1 pruned

All Candidates — Variable Importance
Green = kept (importance > 0) · Grey/red = pruned



Do the two metrics agree?
Upper-right = strong both ways · Upper-left = combo effect only

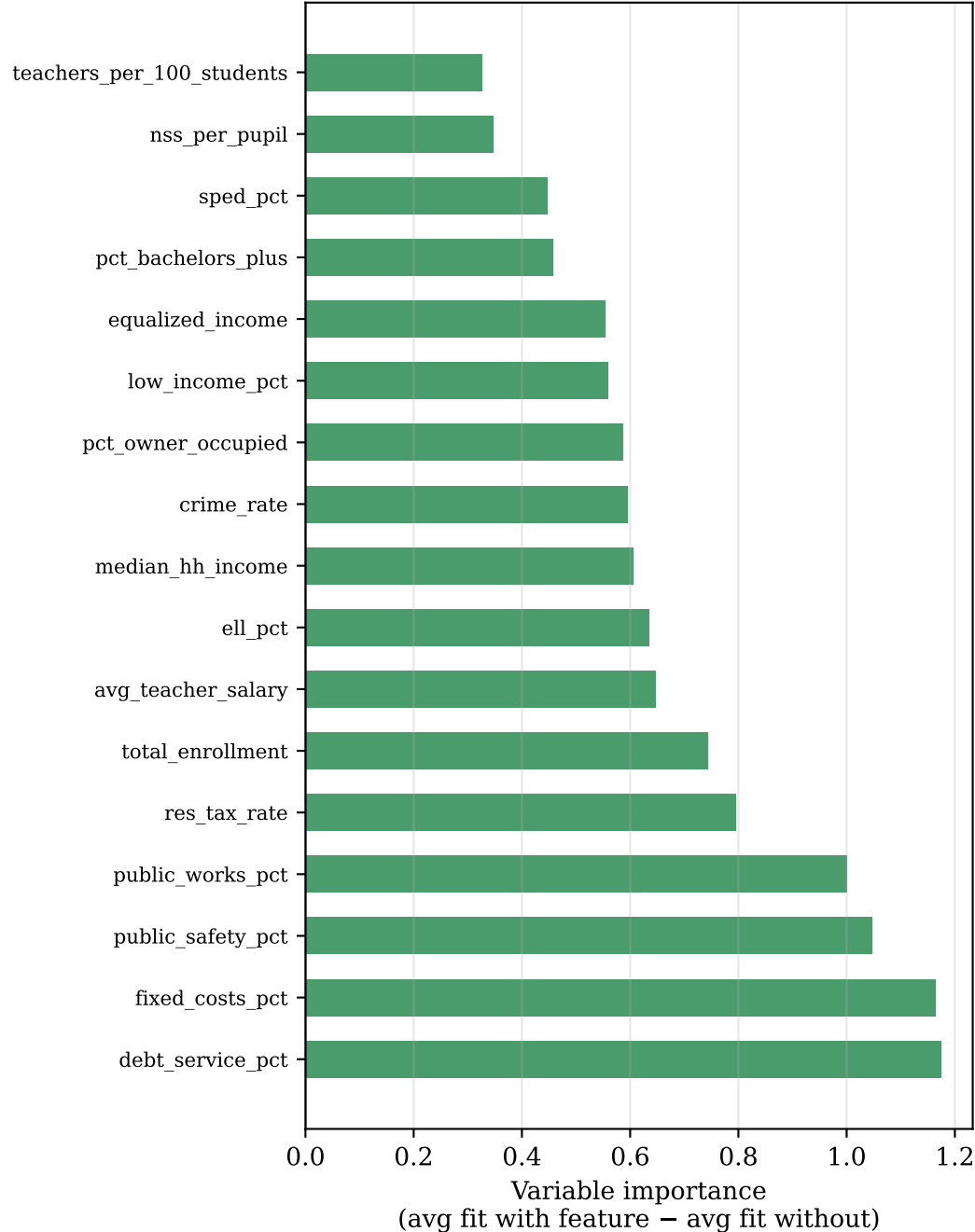


Variable importance: avg adjusted fit of grid cells containing feature – avg fit of cells without it. Univariate LOO r: single-feature leave-one-out correlation.

Saugus RBP Analysis: Education Budget Share

Prediction task = Saugus · Predicted: 36.2pp Actual: 32.3pp Gap: -3.9pp (Fit: 0.2839)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

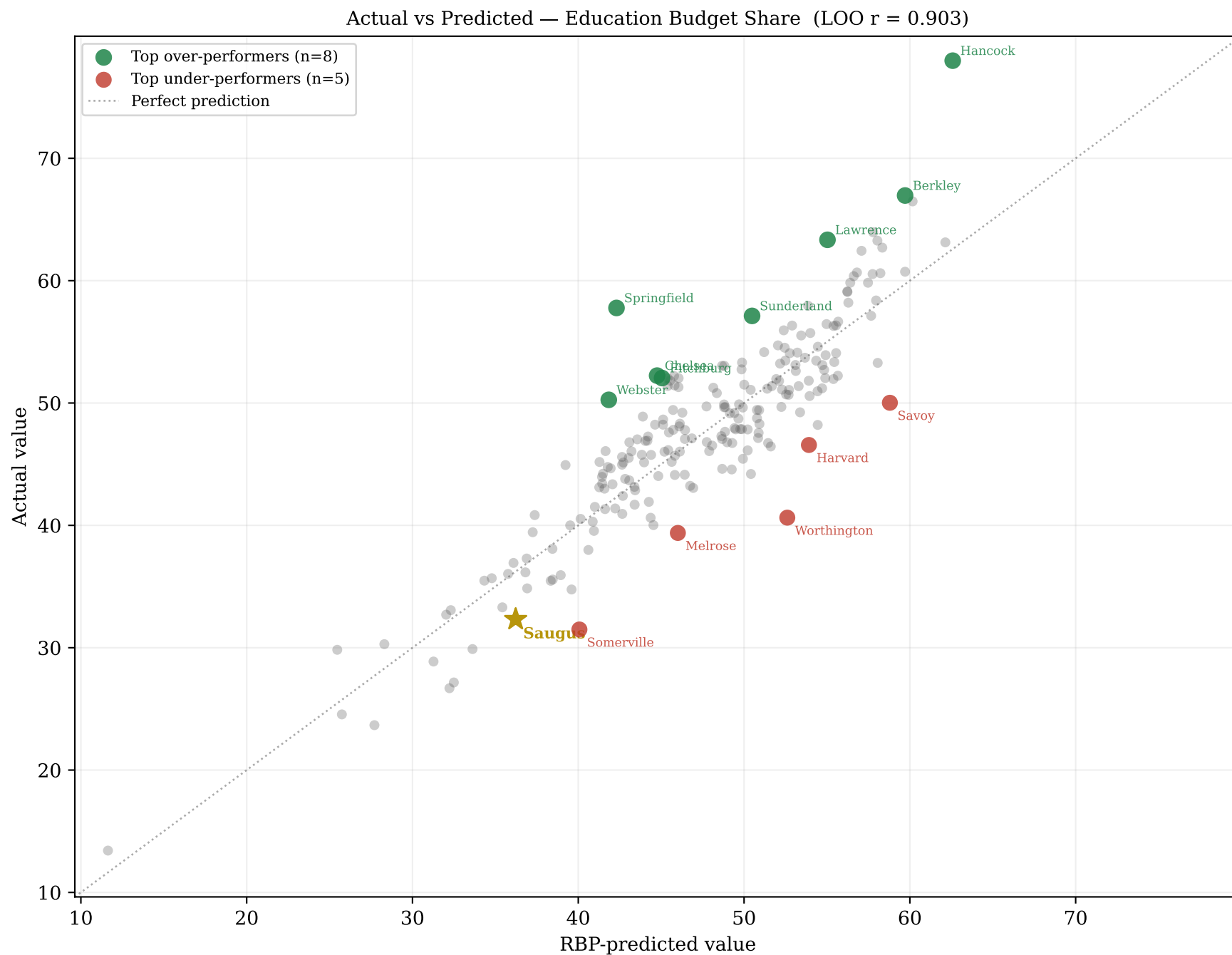
Town	Weight	ed_budget_sh
Quincy	0.0465	28.87
Gloucester	0.0422	33.30
Holbrook	0.0414	39.44
Marlborough	0.0390	43.00
Nahant	0.0331	32.70

Least Relevant Towns (lowest weight)

Town	Weight	ed_budget_sh
Sunderland	-0.0172	57.12
Shutesbury	-0.0164	60.61
Amherst	-0.0148	52.21
Carlisle	-0.0127	60.73
Topsfield	-0.0123	59.83

Over- and Under-Performers: Education Budget Share

Saugus gap: -3.9pp · Gold star = Saugus · Green = over-performers · Red = under-performers



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: Education Budget Share

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Springfie	Hancock	Webster	Imp
debt_service_pct	8.63	3.32	0.00	7.61	+1.174
fixed_costs_pct	25.3	19.7	2.43	19.9	+1.165
public_safety_pct	15.2	11.0	4.61	9.88	+1.047
public_works_pct	7.35	1.35	6.46	2.60	+1.001
res_tax_rate	10.7	16.1	2.79	12.2	+0.796
total_enrollment	3,084	29,468	106.0	1,976	+0.744
avg_teacher_salary	89,778	84,692	59,463	90,071	+0.647
ell_pct	13.0	17.1	0.00	14.5	+0.635
median_hh_income	100,819	51,339	93,125	68,934	+0.606
crime_rate	2,813	8,923	0.00	5,217	+0.596
pct_owner_occupied	73.9	49.5	77.8	58.0	+0.587
low_income_pct	46.6	85.0	36.2	70.3	+0.559
equalized_income	1.17B	2.89B	13M	552M	+0.554
pct_bachelors_plus	31.4	19.6	37.8	24.8	+0.458
sped_pct	18.8	26.8	24.1	25.4	+0.447

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	32.3	36.2	-3.9pp
Springfield	57.8	42.3	+15.5pp
Hancock	78.0	62.6	+15.4pp
Webster	50.3	41.8	+8.4pp
Lawrence	63.3	55.0	+8.3pp
Chelsea	52.2	44.8	+7.5pp
Berkley	67.0	59.7	+7.2pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Education Budget Share

Predicted 36.2pp · Actual 32.3pp · Gap -3.9pp (higher is better) · Saugus ranks 199 of 219 MA districts on this measure

The Bottom Line

Saugus directs 32.3% of its municipal budget to education, 3.9pp below its predicted 36.2% — one of the largest shortfalls in the state (199th of 219 districts, bottom 10%). Unlike the academic measures, this gap reflects a municipal budgeting choice, not a demographic constraint.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Debt service share of town budget	8.6%	5.0%	+1.17
Pension & benefits share of budget	25.3%	17.5%	+1.17
Police & fire share of town budget	15.2%	11.8%	+1.05
Public works share of town budget	7.3%	5.3%	+1.00
Residential tax rate (per \$1,000)	10.7	12.6	+0.80

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Debt service share of town budget	8.6%	2.5%	-6.2pp
Pension & benefits share of budget	25.3%	13.1%	-12.2pp
Police & fire share of town budget	15.2%	10.7%	-4.5pp
Public works share of town budget	7.3%	4.0%	-3.3pp
Residential tax rate (per \$1,000)	10.7	12.3	+1.7

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

All four top local drivers are competing budget categories where Saugus spends more than the state median: debt service (8.6% vs. 5.0%), pensions & benefits (25.3% vs. 17.5%), police & fire (15.2% vs. 11.8%), and public works (7.4% vs. 5.3%). Together these fixed costs crowd out the education share. Of the four outcomes, this is the one where the lever sits with Town Meeting and municipal budgeting, not the school district.

Combined Model Summary: All Four Outcomes

RBP factor selection results across MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, and Education Budget Share

Model	Predicted	Actual	Gap	Direction	LOO r	Features
MCAS Grades 3–8	28.7%	29.0%	+0.3pp	above	+0.946	14
Dropout Rate	3.7%	2.9%	-0.8pp	below	+0.837	15
MCAS Grade 10 (ELA)	42.3%	47.0%	+4.7pp	above	+0.902	14
Education Budget Share	36.2%	32.3%	-3.9pp	below	+0.903	17

Feature Cross-Reference — Which Factors Earn Their Place Across Models

A feature appearing in multiple models is a robust signal, not a model-specific artifact.

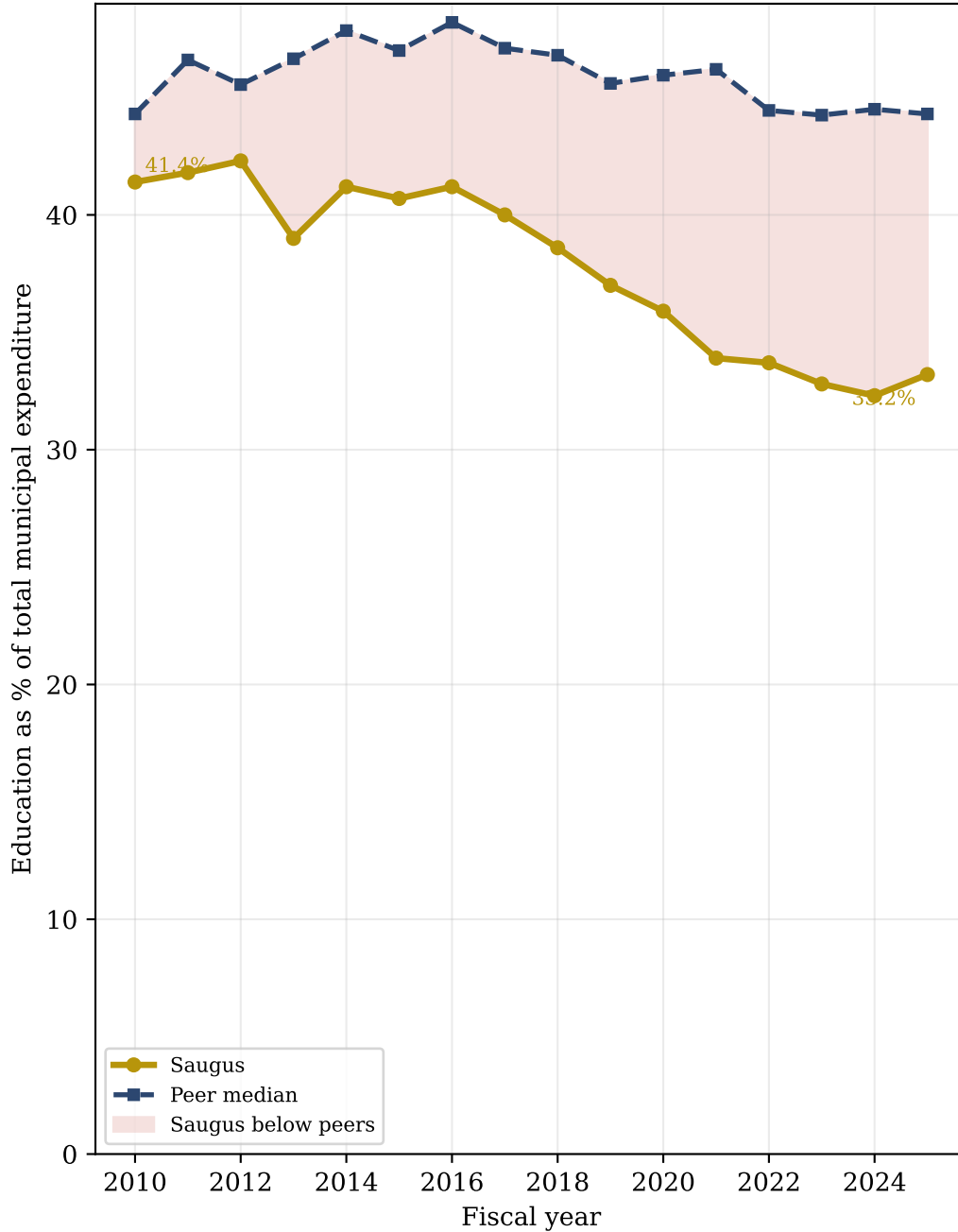
Feature	# Models	MCAS 3-8	Dropout	MCAS 10	Ed Budget
total enrollment	4	+0.168	+0.289	+0.227	+0.744
pct bachelors plus	4	+0.293	+0.362	+0.301	+0.458
equalized income	4	+0.207	+0.330	+0.228	+0.554
median hh income	4	+0.129	+0.405	+0.167	+0.606
avg_teacher_salary	4	+0.115	+0.284	+0.227	+0.647
crime_rate	4	+0.078	+0.371	+0.080	+0.596
ell_pct	4	+0.081	+0.221	+0.134	+0.635
res_tax_rate	4	-0.094	+0.124	+0.034	+0.796
low_income_pct	4	+0.204	+0.185	+0.097	+0.559
pct_owner_occupied	4	+0.092	+0.095	-0.115	+0.587
chronic_absenteeism_pct	3	-0.573	+0.787	-0.342	—
sped_pct	3	-0.261	+0.080	—	+0.447
avg_mcas	2	—	+0.378	+0.232	—
nss_per_pupil	2	—	—	-0.200	+0.348
teachers_per_100_students	2	—	-0.005	—	+0.326
dropout_pct	2	+0.151	—	+0.133	—
debt_service_pct	1	—	—	—	+1.174
fixed_costs_pct	1	—	—	—	+1.165
public_safety_pct	1	—	—	—	+1.047
public_works_pct	1	—	—	—	+1.001
mcas10_ela	1	—	+0.360	—	—
attending_pct	1	-0.191	—	—	—

Green rows = feature appears in 3+ models. Cell colour: green = positive importance, red = negative importance, white = not in that model (—).

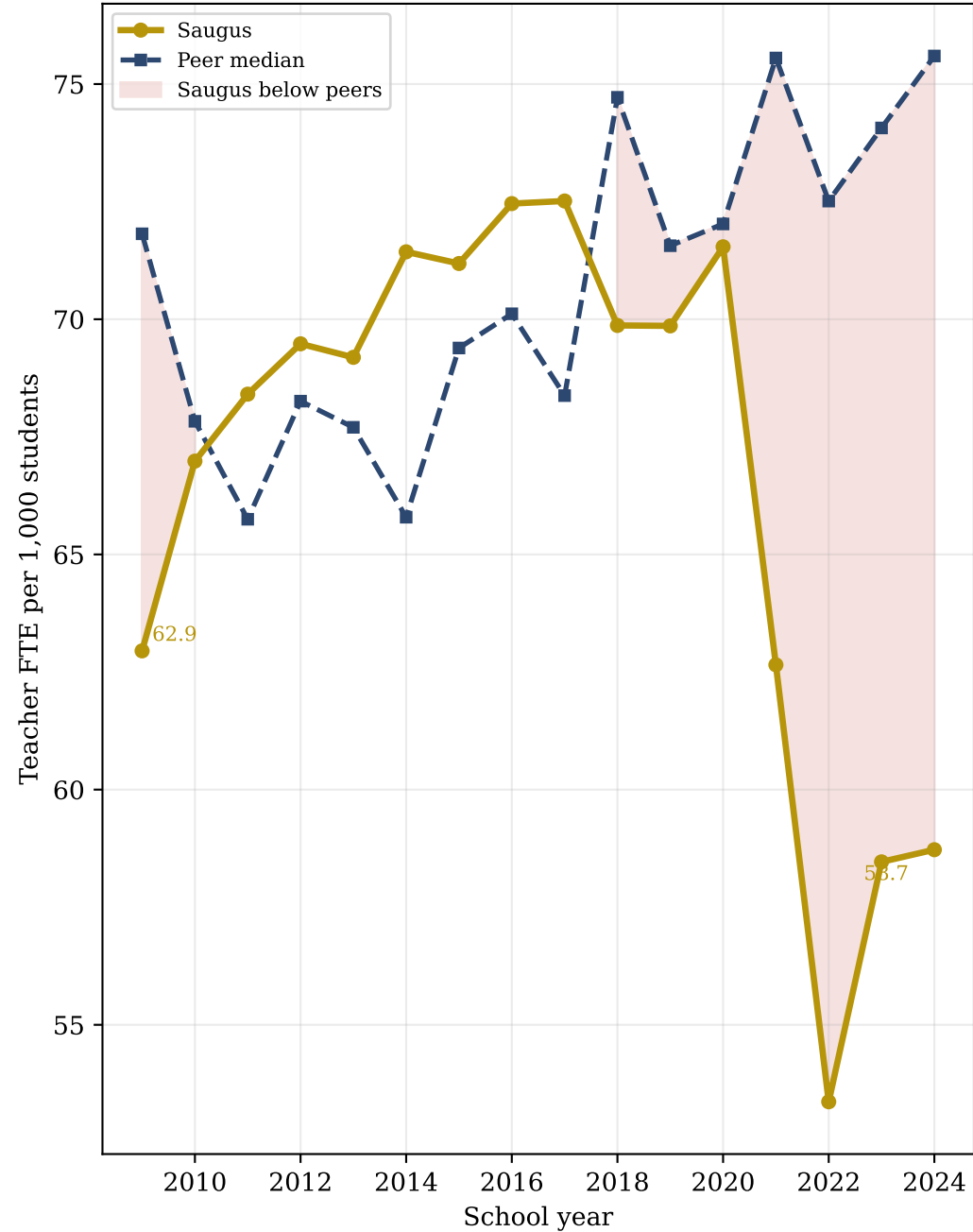
Budget Allocation & Teacher Density — 15-Year Trajectory

Source: MA DLS Schedule A (budget) · MA DESE Staffing (teachers) · Peer comparison: 8 demographically similar overachiever towns

Education Budget Share
(Schedule A, general fund)



Teacher Density
(DESE Staffing + Enrollment)



Independent cross-validation: Ridge regression on 221 MA districts ($R^2=0.83$) identifies chronic absenteeism as the #1 predictor of MCAS outcomes (importance=3.15), confirming the RBP Exhibit 5 finding by a separate method. Peer set: 8 demographically similar overachiever towns (Mahalanobis $d_M \leq 2.0$).

Optimum Profile — What Overachievers Look Like

Each grid shows actionable feature values for Saugus vs overachiever towns. Showing both peer groups tests whether the finding holds across definitions.

Overachievers — all 20 MA districts that beat their demographic prediction

Feature	Saugus	Lawrence	Lenox	Chelsea	Fitchburg	Winchendo	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	29.2%	18.3%	35.4%	34.0%	35.6%	23.8%	-7.4pp	20
Teachers / 100 students	5.9	7.4	10.1	6.7	7.6	6.7	7.5	+1.7	20
Avg teacher salary	\$89,778	\$84,516	\$84,669	\$89,573	\$84,723	\$83,722	\$85,891	\$-3,887	20
Net school spending / pupil	\$17,369	\$23,178	\$23,945	\$22,535	\$20,866	\$19,559	\$20,607	+\$3,238	20
Shown: Lawrence, Lenox, Chelsea, Fitchburg, Winchendon (+ 15 more in median)									

Demographically Similar Overachievers — 18 towns within 2σ Mahalanobis distance of Saugus on income, property wealth, poverty, size, education & ELL

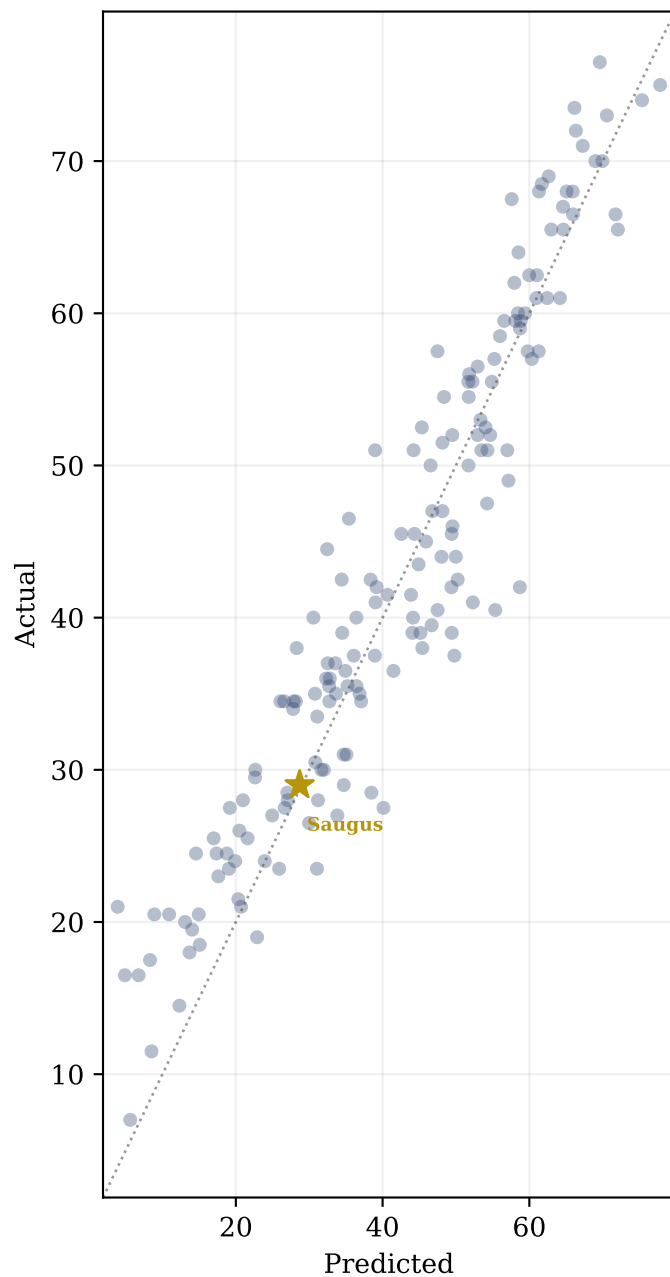
Feature	Saugus	Norwood	Clinton	Oxford	Westport	Fairhaven	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	25.1%	7.7%	19.9%	13.8%	15.6%	21.7%	-9.5pp	18
Teachers / 100 students	5.9	8.8	7.3	7.6	7.5	7.5	7.6	+1.7	18
Avg teacher salary	\$89,778	\$92,324	\$87,059	\$80,053	\$83,017	\$82,224	\$85,891	\$-3,887	18
Net school spending / pupil	\$17,369	\$21,084	\$18,512	\$18,176	\$18,099	\$17,287	\$20,302	+\$2,934	18
Shown: Norwood, Clinton, Oxford, Westport, Fairhaven (+ 13 more in median)									

Similarity metric: Mahalanobis distance $d_M = \sqrt{(\Delta x \cdot \Sigma^{-1} \cdot \Delta x^*)}$, Σ = sample covariance of all ~169 MA districts on (median_hh_income, equalized_income, low_income_pct, total_enrollment, pct_bachelors_plus, ell_pct). Threshold: $d_M \leq 2.0$ (2σ). Pre-specified; not tuned to result. Gap = peer median – Saugus. Red = Saugus below peer median, Green = at or above.

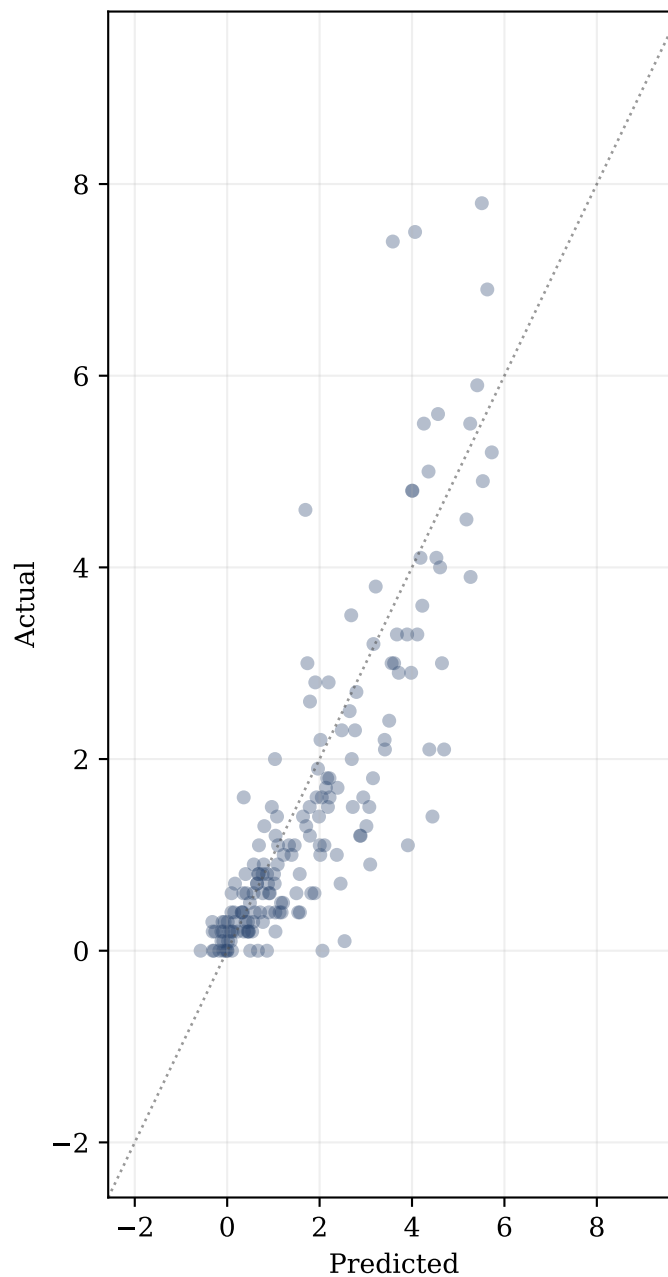
Actual vs. Predicted: All Three Models

Each dot = one MA district. Saugus highlighted in gold.

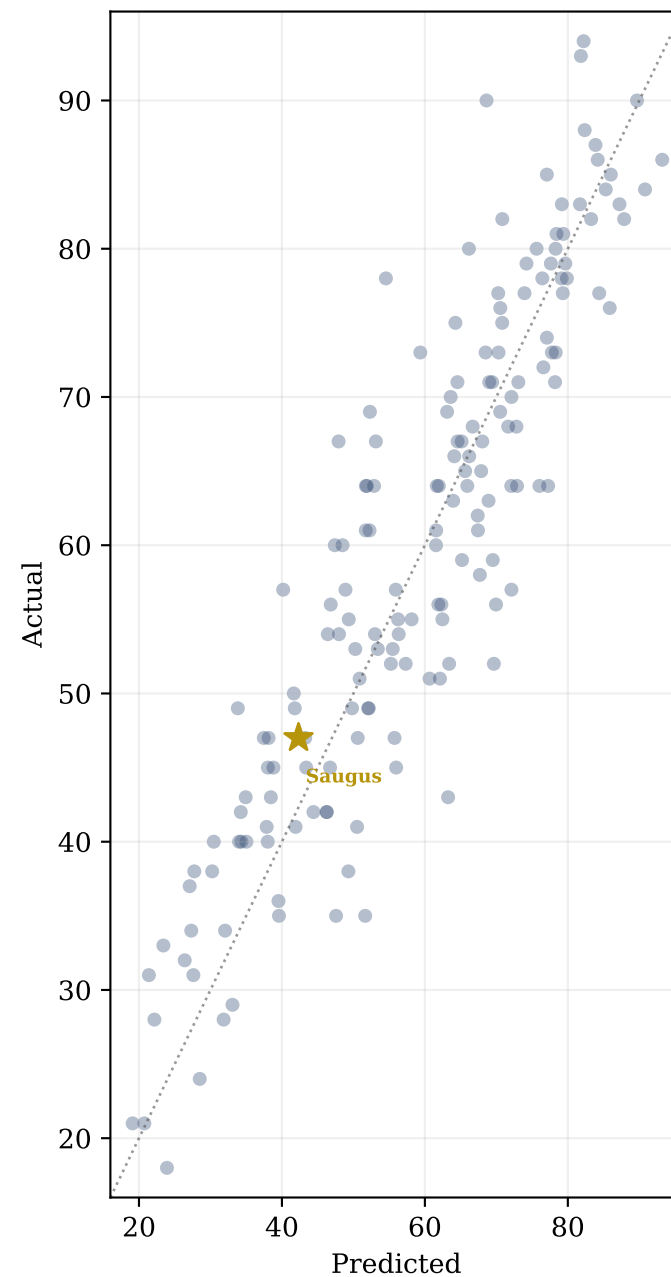
MCAS Grades 3-8
LOO r = 0.946



Dropout Rate
LOO r = 0.837



MCAS Grade 10 (ELA)
LOO r = 0.902



Leave-one-out predictions — each district excluded from its own prediction.